

Class Test I
CITS2200 Data Structures and Algorithms
March 28, 2025

- Total marks for this paper is 10. Please answer by filling the appropriate circle for each question in your answer sheet.
- You should carefully fill your student number in the answer sheet.
- Please return only the answer sheet, **do not** return the question paper.

Q1. Which of the following statements is not true about $O(n \log n)$

(a) $O(n \log n)$ is $O(n^2)$; (b) $O(n \log n)$ is $O(n^3)$; (c) $O(n \log n)$ is $O(\log n)$; (d) $O(n \log n)$ is $O(n^2 \log n)$.

(c)

Q2. You are given two sorted arrays A (with m numbers) and B (with n numbers). You have to write an algorithm for merging these two sorted arrays into a single sorted array. What is the complexity of the best algorithm for solving this problem?

(a) $O(n^2)$; (b) $O(mn)$; (c) $O(m + n)$; (d) $O(mn \log(mn))$.

(c)

Q3. There are four functions, $f(n) = O(n \log n)$, $g(n) = O(n^{\frac{3}{2}})$, $h(n) = O(n)$, and $k(n) = O(\log n)$. When arranged in ascending order, the correct ordering is:

(b)

(a) $h(n), g(n), k(n), f(n)$; (b) $k(n), h(n), f(n), g(n)$; (c) $k(n), h(n), g(n), f(n)$; (d) $g(n), k(n), h(n), f(n)$.

Q4. You have written a Python program for **Insertion Sort** for sorting an array of numbers in ascending order. The algorithm is given an input array that is already sorted in descending order. Which of the following describes the best time complexity of your program on this input?

(a) $O(n)$; (b) $O(n^2)$; (c) $O(n \log n)$; (d) $O(1)$.

(b)

Q7. Which of the following codes is correct for recursively computing the factorial of an integer?

(a) `def factorial(n):
 return n*factorial(n-1)`

(b) `def factorial(n):
 if n==0
 return 2
 else:
 return n*factorial(n-1)`

(c) `def factorial(n):
 if n==0
 return 1
 else:
 return n*factorial(n-1)`

(d) `def factorial(n):
 if n==0
 return 1
 else:
 else return n*factorial(n)`

(c)

Q8. You are given an unsorted array A of length n , and another sorted array B of length k . k is much smaller than n , for example, k may be 100 and n may be 1 million. Your task is to find whether each element in the array A is present in the array B . What is the complexity of the best algorithm for this task?

(a)

(a) $O(n \log k)$; (b) $O(n \log n + k \log n)$; (c) $O(k^2)$; (d) $O(n \log k + k \log k)$.

Q9. There are three independent parts in an algorithm with complexities $O(n^2 \log n)$, $O(n \log n)$ and $O(n)$. What is the overall complexity of the algorithm?

(c)

(a) $O(n \log n)$; (b) $O(n^2)$; (c) $O(n^2 \log n)$; (d) $O(n^2 \log n)$.

Q10. Given an unsorted array of n numbers, what is the complexity of the best algorithm for finding the sum of the numbers in the array?

(a)

(a) $O(n)$; (b) $O(n \log n)$; (c) $O(n^2)$; (d) $O(n^2 \log n)$.

End of Paper